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POWER BI

Complete Exam Study Guide

Coursera Audit Course | Data Analyst (PL-300 aligned) | MCQ Exam

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Exam Details	
Exam dates	10th and 11th October
Shift 1	9:00 AM - 10:00 AM
Shift 2	10:00 AM - 11:00 AM
Shift 3	12:00 PM - 1:00 PM
Format	MCQs - short, basic factual recall (UI paths, definitions, lists, storage modes)
Marks	Check your college notice for total marks and passing criteria

Module	Course Schedule - What is covered
M1	Data preparation and ETL - sources, connectors, storage modes, refresh
M2	Power Query transformations and data modeling - schemas, joins, profiling
M3	DAX and measures - columns, measures, context, time intelligence
M4	Visualizations and reports - charts, KPIs, slicers, drill, bookmarks

Built from 30,000+ words of course material over 5 rounds of processing - the most important and handy guide in one PDF.

Contents

Contents	2
Legend - What the Coloured Boxes Mean	3
Module 1 - Data Preparation and ETL	4
Module 2 - Power Query and Data Modeling	8
Module 3 - DAX and Measures	13
Module 4 - Visualizations and Reports	17
Final Cheat Sheet and Exam Tips	21
Bonus - Excel and PowerPoint Shortcuts	22
Mock Exam Drill - 50 Questions	23
Mock Drill - Answer Key	29

How this guide is organised | DEFINITION TO REMEMBER

- Legend page > Module 1 to 4 (each: syllabus table, notes, MCQs with answers, exercises with answers)
- Then: Final Cheat Sheet and Tips > Critical Facts tables > Excel and PowerPoint shortcuts > 50-question Mock Drill with answers > Last page
- Each module ends with a cheat sheet style recap and important questions with solutions.

Legend - What the Coloured Boxes Mean

RED BOX | VERY IMPORTANT - HIGH-CHANCE MCQ

- Very important / high-chance MCQ topic. If you only have 10 minutes, read the red boxes.

TEAL BOX | COMMON CONFUSION EXPLAINED

- Common confusions explained - pairs that look alike (Append vs Merge, Drill down vs Drill through, Measure vs Calculated column).

YELLOW BOX | DEFINITION TO REMEMBER

- Definitions to remember - short, exam-style one-liners.

Inline highlight: **yellow highlighted bold text** inside a sentence = the exact keyword the MCQ will ask for.

Priority marks used in tables: **HIGH** = very likely, **MED** = likely, **LOW** = unlikely but possible. Green tables = MCQ answer keys. Purple tables = match-the-following exercises.

Study plan for exam day | VERY IMPORTANT - HIGH-CHANCE MCQ

- Read all RED boxes first, then TEAL boxes (confusions cost marks).
- Attempt each module's MCQs, check answers, re-read the matching notes for any you missed.
- Finish with the 50-question Mock Drill - aim for 40+ correct.

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Module 1 - Data Preparation and ETL

Syllabus Table (Blueprint)

Topic	What you must know	Priority
ETL	Extract, Transform, Load - the 3 stages	HIGH
Data sources	Flat files, relational, NoSQL, blob storage	HIGH
Data structure types	Structured, unstructured, semi-structured	HIGH
Connectors	100+ connectors; Beta and Preview	MED
Storage modes	Import, DirectQuery, Dual (+ composite model)	HIGH
Refresh	Schedule Refresh, triggers and actions, semantic model	HIGH
PL-300	Microsoft Power BI Data Analyst exam	LOW

1.1 Power BI and ETL

Power BI | DEFINITION TO REMEMBER

- Microsoft tool to **connect, transform, visualize and analyze** data. It simplifies combining many data sources into one report.

ETL = Extract, Transform, Load | VERY IMPORTANT - HIGH-CHANCE MCQ

- Extract**: collect data from different sources (and choose storage modes).
- Transform**: clean, merge and profile data so it is ready for analysis.
- Load**: load transformed data into Power BI / the analysis tool (profiling, advanced queries, referencing queries, data flows, Advanced Editor are discussed here).
- Order is always **E - T - L**.

1.2 Data Sources

Category	Examples	Key point
Flat file	CSV, TXT, Fixed width, Excel workbook	Single table, uniform structure, no hierarchies. If file location changes, refresh errors occur.
Relational	SQL Server, MySQL, Oracle	High reliability, integrity and security. SQL Server suits large organizations.
NoSQL	MongoDB, Cassandra	Store large unstructured / semi-structured data.
Blob	Azure Blob Storage	Cloud storage for images, videos, documents. Blob = Binary Large Object.

Structured vs Unstructured vs Semi-structured | COMMON CONFUSION EXPLAINED

- Structured** = relational, rows and columns, searchable (Excel, SQL). Adding fields is difficult.
- Unstructured** = no predefined format: text docs, audio/video, social media, images. Non-relational.
- Semi-structured** = not fully organised; uses **tags** for hierarchy (e.g. video metadata). Data serialization converts it to a specific format for storage/transmission.

1.3 Connectors

- Connectors are links that carry data between a source and an application like Power BI.
- 100+ connectors**; files, databases and web services are the most used. Free to use; about 1-2 new ones added monthly.
- Beta** and **Preview** connectors have limited support - **not for production** use.

1.4 Storage Modes

Import vs DirectQuery vs Dual | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Import**: data imported into **memory**, compressed, quick access; needs refresh to update. Retrieves all data.
- **DirectQuery**: connects **directly to the source**; real-time queries; can impact performance. Retrieves only metadata, column names and data types.
- **Dual**: combines both; the **Power BI service picks** the most efficient mode per query.
- **DirectQuery/Dual to Import is IRREVERSIBLE** (a warning appears; click OK).
- **Composite model** = a model using multiple sources or multiple storage modes; set the mode per table. Best performance with Import/Dual tables.

How to see/change the storage mode | COMMON CONFUSION EXPLAINED

- Model view icon (left side) shows tables. Header colour: **no colour = Import; same colour = DirectQuery**.
- Right-click table > **Properties** > expand Properties pane > **Advanced** section > **Storage Mode** dropdown.
- For a new connection (e.g. Azure SQL): **Data Connectivity Mode** section > Import or DirectQuery (Import is chosen most of the time).

1.5 Getting Data - Step Recaps

Setting up a flat (Excel) file source | VERY IMPORTANT - HIGH-CHANCE MCQ

- Home tab > **Get Data** (or its down arrow > **More** to open the Get Data dialog) > select **Excel** > Connection window: choose workbook > **Open**.
- **Navigator** window: tick the table checkbox (preview) > **Load** (or Transform Data).
- If file location changes: **File > Options and Settings > Data Source Settings** > update connection strings; **Edit or Clear permissions**. Reconnect to the **same structure** - structural changes break the model.

Connecting to SQL (e.g. Azure SQL) | DEFINITION TO REMEMBER

- Home > Get data > search **Azure SQL** > Connect > enter server name and database > choose Import or DirectQuery.
- **3 sign-in options**: Windows account, database credentials, Microsoft account (Azure AD). Navigator > select tables > Load or Transform.
- Permissions = privacy level + credentials. Connection strings = settings that say how to connect.

1.6 Refresh, Triggers and Actions (Power BI Service)

Schedule Refresh path | VERY IMPORTANT - HIGH-CHANCE MCQ

- Power BI Service > **workspace** > select the **semantic model** > top **Refresh** drop-down > **Schedule Refresh**.
- In **Refresh** section: verify time zone > switch the refresh schedule **On** > frequency (e.g. **Daily**) > time (e.g. 5 AM) > add email contacts for failure notifications > **Apply**.
- **Trigger** = event/condition that starts a workflow (new data or schedule). **Action** = the task (refresh report, send alert, update dataset).

Semantic model vs Dataset | COMMON CONFUSION EXPLAINED

- They are the **same thing** - 'semantic model' is the newer name in Power BI Service. It organizes raw data into meaningful categories and relationships and powers reports.

Importing into Power Query | DEFINITION TO REMEMBER

- Home > Get Data > choose source > configure connection > **Transform Data** (opens Power Query Editor in a separate window) > apply transformations > load.
- **PL-300** = exam for Microsoft Power BI Data Analyst certification: prepare data, model data, visualize/analyze, deploy/maintain.

Module 1 - Quick Cheat Sheet

- ETL = Extract, Transform, Load. Flat files = CSV, TXT, Fixed width, Excel. Relational = SQL Server, MySQL, Oracle. NoSQL = MongoDB, Cassandra.
- Storage modes = Import / DirectQuery / Dual. Irreversible: DirectQuery or Dual to Import.
- Schedule Refresh = Service > semantic model > Refresh > Schedule Refresh. Data Source Settings = File > Options and Settings.

Practice MCQs - Module 1

Q1. What does ETL stand for?

- (A) Extract, Transform, Load
- (B) Export, Translate, Link
- (C) Extract, Transfer, Load
- (D) Evaluate, Test, Launch

Q2. Which storage mode imports compressed data into memory?

- (A) Import
- (B) Composite
- (C) Dual
- (D) DirectQuery

Q3. Which storage mode queries the source in real time?

- (A) DirectQuery
- (B) Snapshot
- (C) Dual
- (D) Import

Q4. In Dual mode, who selects the most efficient mode per query?

- (A) The report user
- (B) The Power BI service
- (C) The database admin
- (D) Power Query Editor

Q5. Which change of storage mode is irreversible?

- (A) Import to DirectQuery
- (B) Import to Dual
- (C) None of them
- (D) DirectQuery or Dual to Import

Q6. Which is NOT a flat file type?

- (A) CSV
- (B) MongoDB
- (C) Fixed width
- (D) TXT

Q7. Data that uses tags to organise itself is called:

- (A) Unstructured data
- (B) Structured data
- (C) Semi-structured data
- (D) Relational data

Q8. Where is Schedule Refresh configured?

- (A) Power BI Service, semantic model > Refresh drop-down
- (B) Data Source Settings
- (C) Model view Properties
- (D) Power Query Editor Home tab

Q9. Which path opens Data Source Settings?

- (A) Insert > Buttons
- (B) File > Options and Settings > Data Source Settings
- (C) View > Sync slicers
- (D) Home > Transform Data

Q10. Beta and Preview connectors should be used:

- (A) Only in Power BI Service
- (B) For all production reports
- (C) Not for production (limited support)
- (D) Only with DirectQuery

Answer Key - Module 1

Q	Ans	Answer and why
Q1	A	Extract, Transform, Load - ETL is the 3-stage process.
Q2	A	Import - Import = in-memory, quick access.
Q3	A	DirectQuery - DirectQuery connects directly to the source.
Q4	B	The Power BI service - Dual combines Import and DirectQuery.
Q5	D	DirectQuery or Dual to Import - A warning message appears first.
Q6	B	MongoDB - MongoDB is a NoSQL database.
Q7	C	Semi-structured data - Tags give hierarchy.
Q8	A	Power BI Service, semantic model > Refresh drop-down - Refresh > Schedule Refresh.
Q9	B	File > Options and Settings > Data Source Settings - Use it to update connection strings and permissions.
Q10	C	Not for production (limited support) - They have limited support/functionality.

Exercise - Match the Following

Column A	Column B
1. Trigger	A. Cloud storage for unstructured data
2. Action	B. Event that starts a workflow
3. Azure Blob Storage	C. Task such as refresh report or send alert
4. Composite model	D. Another name for dataset in Service
5. Semantic model	E. Multiple sources or storage modes

Exercise - Fill in the Blanks

- The three ETL stages are Extract, _____ and Load.
- Header with no colour in Model view means _____ mode.
- _____ mode has real-time queries to the source.
- Flat file types: CSV, TXT, Fixed width and _____.
- Relational sources: SQL Server, MySQL and _____.

Exercise answers | DEFINITION TO REMEMBER

- Match: 1-B, 2-C, 3-A, 4-E, 5-D
- Fill: 1. Transform | 2. Import | 3. DirectQuery | 4. Excel workbook | 5. Oracle

Module 2 - Power Query and Data Modeling

Syllabus Table (Blueprint)

Topic	What you must know	Priority
Power Query Editor	Tabs, Applied Steps, queries pane	HIGH
Cleaning	Filter, sort, promote headers, null vs empty, remove duplicates, data types	HIGH
Pivot / Unpivot	Wide to narrow and back	HIGH
Append / Merge / Joins	Add rows vs add columns; 4 join types	HIGH
Data profiling	Column quality, distribution, profile	HIGH
Statistics	Distribution, outliers, standard deviation	MED
Schemas	Flat, Star, Snowflake; fact and dimension tables	HIGH
Relationships	Cardinality, cross-filter direction	HIGH

2.1 Power Query Editor Basics

Power Query | DEFINITION TO REMEMBER

- Data transformation and preparation tool in Power BI Desktop. **Power Query Editor** = the graphical interface to design and manage queries. Features: data connectivity, extraction and transformation, query reusability (applied steps), performance and scalability.

Editor part	Location / use
Ribbon tabs	Home, Transform, Add Column, View
Queries pane	Left side - lists queries
Data preview pane	Centre - preview after transformations
Applied Steps	Right pane, below ribbon - history of transformations

Applied Steps List - icons | VERY IMPORTANT - HIGH-CHANCE MCQ

- Select a step = view data at that state. **X icon** = removes that step **and all later (dependent) steps**.
- Gear icon** = open settings to modify the step. **Drag** = reorder. **Right-click > Rename** = rename.
- Hold **Ctrl** and click column headers to select multiple columns; right-click > Remove Columns / Rename.

2.2 Cleaning and Shaping

- Filtering** = show only rows meeting criteria (column header drop-down arrow: text/number/date filters, empty, null).
- Sorting** = Home tab > Sort group: **Sort Ascending A to Z / Sort Descending Z to A**.
- Promote header** = Home tab > Transform group > **Use First Row as Headers**.
- Remove duplicates** = Home > **Remove Rows** > **Remove Duplicates** (keeps one of each identical group).
- Fix data type** (e.g. Order Date) = select column > **Transform** tab > **Data Type** > Date.
- Replace Values** = Transform tab > Replace Values > fill Value to Find and Replace With > OK.
- Close and Apply** (Home) = apply changes and return to Power BI Desktop.

Null vs Empty | COMMON CONFUSION EXPLAINED

- Null** = missing value in **numeric or date** columns. **Empty** = blank cells in **text** columns.

Data cleaning vs Data transformation | COMMON CONFUSION EXPLAINED

- Data cleaning** = preliminary examination to find incorrect data/inconsistencies (a subset). **Data transformation** = broader: cleaning, merging, profiling. Example inconsistency: 'where' vs 'warehouse'.

2.3 Pivot, Unpivot, Append, Merge

Unpivot vs Pivot | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Unpivot** = wide to narrow: column headers become row values. Transform tab > **Unpivot** (e.g. 2022 and 2023 columns become Attribute/Value; rename to Year and Target amount).
- **Pivot** = narrow to wide: rows become columns. Transform tab > **Pivot Column** > Values column > Advanced options > Aggregate function (e.g. **Count All**).
- **Wide format** = many columns (a column per region). **Narrow format** = few columns, related info stacked.

Append vs Merge | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Append** = adds **rows** from one or more tables. Needs equal columns with **same names and data types**; extra columns get **nulls**. Home tab > **Append Queries** (into existing) or **Append Queries as New** (new master table).
- **Merge** = combines **columns** from multiple tables through a **shared (join key) column**.
- **Join** = combining data from different places into a bigger dataset.

Join type	Result
Left outer	All rows from left + matching from right; nulls for unmatched right
Right outer	All rows from right + matching from left; nulls for unmatched left
Full outer	All records from both tables regardless of matching
Inner	Only matching rows from both tables

2.4 Data Profiling and Statistics

3 profiling tools (View tab > Data Preview group) | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Column quality** = % valid, error and empty rows per column.
- **Column distribution** = frequency and distribution visuals (hover for distinct and unique counts).
- **Column profile** = statistics (min, max, average, standard deviation) + value distribution chart for the selected column.

Unique vs Distinct | COMMON CONFUSION EXPLAINED

- **Unique values** = values that appear **only once**. **Distinct values** = number of **different** values regardless of how many times each appears.
- **Distribution** = how data points spread in a dataset. **Outlier** = point far from other observations; can skew averages. Min-max scaling/normalization handles outliers.
- **Standard deviation** = measure of variation/dispersion. Mean = average, Mode = most frequent, Min/Max = smallest/largest.

2.5 Schemas

Schema	Structure	Remember
Flat	All attributes in a single table	Simplest; suited to small data; redundancy, slow on large data
Star	Central fact table + dimension tables (star shape)	Denormalized; fast queries; less flexible
Snowflake	Star with dimension tables normalized into more related tables	Less redundancy; more complex, slower queries

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Fact table vs Dimension table | COMMON CONFUSION EXPLAINED

- **Fact** = measurements/metrics (Order ID, Product ID, Quantity, Total Price); sits at the **centre**.
- **Dimension** = descriptive text attributes (customer, product, date); radiates like **star points**.
- **Normalization** = split tables into more related tables (snowflake). **Denormalization** = combine tables (star).

2.6 Relationships

Cardinality and Cross-filter direction | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Cardinality** = how tables relate (one-to-many, many-to-one). One-to-many is common in star schema. Set in the **Create Relationship** dialog.
- **Cross-filter direction**: **Single** (default, one way) vs **Bi-directional** (both ways, can hurt performance).
- Star schema set-up: **File > Options and Settings > Options > Data Load > untick 'Auto detect new relationships'** > load tables > Model view > **Manage Relationships** > New > pick key columns > Cardinality **Many to one** > Filter direction **Single**.
- Snowflake: create new category/subcategory tables with **Table Tools > New table** (DAX).

Module 2 - Quick Cheat Sheet

- Applied Steps = right pane; X removes step + later steps; Gear edits. Unpivot wide>narrow; Pivot narrow>wide. Append = rows; Merge = columns.
- Profiling: Column quality / distribution / profile. Schemas: Flat (1 table), Star (fact + dims), Snowflake (normalized dims). Default filter direction = Single.

Practice MCQs - Module 2

Q1. Where is the Applied Steps list in Power Query Editor?

- (A) Left pane
- (B) Bottom status bar
- (C) Inside the View tab
- (D) Right pane, below the ribbon

Q2. Clicking the X icon beside an applied step does what?

- (A) Moves it up
- (B) Renames the step
- (C) Opens its settings
- (D) Removes that step and all subsequent dependent steps

Q3. Which operation converts column headers into row values?

- (A) Unpivot
- (B) Pivot
- (C) Append
- (D) Merge

Q4. Append requires the tables to have:

- (A) Only numeric columns
- (B) A shared join key
- (C) Different column names
- (D) Equal columns with same names and data types

Q5. Which join keeps all rows from the left table plus matches from the right?

- (A) Right outer
- (B) Inner
- (C) Left outer
- (D) Full outer

Q6. Which schema has a central fact table linked to dimension tables?

- (A) Snowflake schema
- (B) Star schema
- (C) Flat schema
- (D) Composite schema

Q7. Which profiling tool gives min, max, average and standard deviation?

- (A) Column distribution
- (B) Column quality
- (C) Column header
- (D) Column profile

Q8. Default cross-filter direction in Power BI is:

- (A) Bi-directional
- (B) Single
- (C) None
- (D) Both

Q9. Where is 'Auto detect new relationships' turned off?

- (A) Home > Transform Data
- (B) Options > Data Load
- (C) Model view > Properties
- (D) View > Sync slicers

Q10. Blank cells in a TEXT column are called:

- (A) Distinct
- (B) Null
- (C) Outlier
- (D) Empty

Answer Key - Module 2

Q	Ans	Answer and why
Q1	D	Right pane, below the ribbon - Queries pane is on the left.
Q2	D	Removes that step and all subsequent dependent steps - Gear icon edits it.
Q3	A	Unpivot - Wide to narrow.
Q4	D	Equal columns with same names and data types - Extra columns get nulls.
Q5	C	Left outer - Unmatched right values are null.
Q6	B	Star schema - Star shape.
Q7	D	Column profile - Quality = valid/error/empty.
Q8	B	Single - Bi-directional can impact performance.
Q9	B	Options > Data Load - File > Options and Settings > Options.
Q10	D	Empty - Null is for numeric/date columns.

Exercise - Match the Following

Column A	Column B
1. Outlier	A. Descriptive attributes
2. Fact table	B. Normalized dimension tables
3. Dimension table	C. Point deviating strongly from others
4. Snowflake	D. Only matching rows from both tables
5. Inner join	E. Metrics of a business process

Exercise - Fill in the Blanks

1. Merge combines tables using a shared _____ column.
2. Column _____ shows valid, error and empty rows.
3. Pivot converts rows to columns - narrow to _____.
4. The default filter direction is _____.
5. Distinct values count _____ values regardless of repeats.

Exercise answers | DEFINITION TO REMEMBER

- Match: 1-C, 2-E, 3-A, 4-B, 5-D
- Fill: 1. join key | 2. quality | 3. wide | 4. Single | 5. different

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Module 3 - DAX and Measures

Syllabus Table (Blueprint)

Topic	What you must know	Priority
DAX basics	Formula, functions, operators, VAR/RETURN	HIGH
Calculated column / table / measure	Differences and where to create	HIGH
Row vs filter context	Definitions and order	HIGH
CALCULATE	Changes filter context; modifiers	HIGH
Quick and custom measures	No-code vs DAX	HIGH
Time intelligence	TOTALYTD, SAMEPERIODLASTYEAR, DATEADD, PARALLELPERIOD...	HIGH
Statistical functions	AVERAGE, MEDIAN, COUNT, DISTINCTCOUNT, MIN, MAX	MED
BI concepts	Visualization, analysis, predictive analytics	LOW

3.1 DAX Basics

DAX | DEFINITION TO REMEMBER

- **Data Analysis Expressions** - syntax for creating calculations in Power BI. A **function** = name + parentheses with parameters (write names in CAPITALS). Operators are the building blocks of a formula.
- **VAR** defines a variable; **RETURN** gives the expression's result after the variables.
- Table names in single quotes, column names in square brackets, e.g. 'Sales'[Quantity].
- Typing **=** triggers the list of suggested functions; press **Enter** to execute the formula.

Calculated column vs Measure vs Calculated table | COMMON CONFUSION EXPLAINED

- **Calculated column** = new column in a table via DAX; **static, stored** in the table; row by row. Create: **Table Tools > New Column** (Calculations group).
- **Measure** = aggregation on model fields; **dynamic**, calculated in the visual's context, not stored; reusable. Create: **Table Tools > New Measure**.
- **Calculated table** = new table derived from existing tables with a DAX formula (**New table**). Useful for summary tables.
- Format after creating: Profit as **currency**, Profit Margin as **percentage**. Profit margin = (Sales revenue - COGS) / sales revenue.

3.2 Context

Row context vs Filter context | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Row context** = the **current row** being evaluated (used in calculated columns). Does NOT automatically propagate through relationships.
- **Filter context** = filter constraints applied **before** DAX evaluation; decides which rows are included. Multiple filters on same table = logical **AND**. Propagates via relationships (per cross-filter direction).
- Evaluation order: **filter context first, then row context**.

3.3 CALCULATE

CALCULATE | VERY IMPORTANT - HIGH-CHANCE MCQ

- Evaluates an expression in a context **modified by specified filters**. First argument = expression, then filters. Existing filters are **overridden**.
- Modifiers: **KEEPFILTERS** (keep existing filters), **CROSSFILTER** (cross-filter direction), **USERELATIONSHIP** (which relationship to use).
- Examples: Bike Sales = CALCULATE(Total Sales, Category = Bikes); Sales of Blue Products in USA adds Colour and Country filters.
- **SUMX** iterates row by row: multiplies columns and sums (Total Sales = quantity x unit price). **TOPN** + number gives top products (Top Two Products).

3.4 Quick vs Custom Measures

Quick measure vs Custom measure | COMMON CONFUSION EXPLAINED

- **Quick measure** = pre-built; runs DAX behind the scenes, **no code needed**. Path: Report/Data view > Calculations group > **Quick measure** (or Data pane ellipses > New Quick measure) > select calculation and fields > Add.
- **Custom measure** = user-defined DAX (e.g. SUMX, TOPN). Give unique names.
- Quick measure categories (6): **Aggregate per category, Filters, Time intelligence, Totals, Mathematical operations, Text**.
- Example: SalesYTD via 'Year-to-date total' with Sales as base value and Order Date as date.

3.5 Time Intelligence and Statistical Functions

Function	What it does
TOTALYTD	Sums values from Jan 1 to a chosen date
DATESYTD	Table of dates from start of year to date; used inside CALCULATE
DATESBETWEEN	Dates between two specified dates (e.g. SummerSales)
DATESINPERIOD	Dates in a period from a start date (e.g. last 30 days)
SAMEPERIODLASTYEAR	Same period last year - year-over-year comparison
DATEADD	Shifts dates forward/backward by an interval (e.g. one month)
PARALLELPERIOD	Shifts by a defined number of intervals; more flexible (e.g. two years back)

- Abbreviations: **YTD** Year-to-Date, **MTD** Month-to-Date, **YOY** Year-over-Year.
- Auto date/time creates a hidden date table per date column; use custom DAX for complex needs.
- Statistical functions (6): **AVERAGE, MEDIAN, COUNT, DISTINCTCOUNT, MIN, MAX**. AVERAGEX = average over each row; SUMMARIZE = grouped summary table.

Business Intelligence (BI) in one go | DEFINITION TO REMEMBER

- **BI** converts raw data into meaningful, actionable information. **Data visualization** = graphical representation (charts, graphs, maps). **Data analysis** = deeper dive into the 'why'. **Predictive analytics** = forecasts using current/historical data.
- Chart picks: **Column** vertical (<10 categories), **Bar** horizontal (long labels), **Line** trends, **Area** part-to-whole over time, **Pie** proportions, **Slicers** visual filters.

Module 3 - Quick Cheat Sheet

- Column = static, row context. Measure = dynamic, filter context. CALCULATE = modifies filter context. VAR ... RETURN.
- New Column / New Measure = Table Tools > Calculations group. Quick measure = 6 categories, no code.

Practice MCQs - Module 3

Q1. Which statement about a calculated column is true?

- (A) It is created in Power Query only
- (B) It is stored in the table and evaluated row by row
- (C) It is dynamic like a measure
- (D) It cannot be used in visuals

Q2. Where do you create a New Measure?

- (A) Table Tools tab > Calculations group
- (B) View tab > Sync slicers
- (C) Insert tab > Buttons
- (D) Home tab > Append Queries

Q3. Which function modifies the filter context?

- (A) TOPN
- (B) CALCULATE
- (C) SUMX
- (D) AVERAGE

Q4. Row context refers to:

- (A) Filters applied before evaluation
- (B) Relationships between tables
- (C) The report page
- (D) The current row being evaluated

Q5. Which function sums values from January 1 to a chosen date?

- (A) PARALLELPERIOD
- (B) DISTINCTCOUNT
- (C) TOTALYTD
- (D) DATEADD

Q6. Which function returns the same period last year?

- (A) SAMEPERIODLASTYEAR
- (B) KEEPFILTERS
- (C) DATESINPERIOD
- (D) DATESBETWEEN

Q7. Which CALCULATE modifier prevents existing filters from being overridden?

- (A) CROSSFILTER
- (B) KEEPFILTERS
- (C) USERELATIONSHIP
- (D) VAR

Q8. Which keyword gives the result after variables in DAX?

- (A) RETURN
- (B) VAR
- (C) SUMX
- (D) TOPN

Q9. How many quick measure categories are listed in the course?

- (A) Four
- (B) Eight
- (C) Six
- (D) Three

Q10. Which function counts the number of different values?

- (A) MEDIAN
- (B) SUMMARIZE
- (C) COUNT
- (D) DISTINCTCOUNT

Answer Key - Module 3

Q	Ans	Answer and why
Q1	B	It is stored in the table and evaluated row by row - Measures are dynamic.
Q2	A	Table Tools tab > Calculations group - Same group as New Column.
Q3	B	CALCULATE - It evaluates an expression with modified filters.
Q4	D	The current row being evaluated - Filter context is applied first.
Q5	C	TOTALYTD - YTD = Year-to-Date.
Q6	A	SAMEPERIODELASTYEAR - Used for YoY comparison.
Q7	B	KEEPFILTERS - Keeps active filters.
Q8	A	RETURN - VAR defines, RETURN outputs.
Q9	C	Six - Aggregate, Filters, Time intelligence, Totals, Math, Text.
Q10	D	DISTINCTCOUNT - COUNT counts rows.

Exercise - Match the Following

Column A	Column B
1. SUMX	A. Chooses which relationship to use
2. TOPN	B. Iterates rows, multiplies and sums
3. DATEADD	C. Shifts dates by an interval
4. USERELATIONSHIP	D. Returns top N products
5. Quick measure	E. No-code DAX calculation

Exercise - Fill in the Blanks

1. A _____ column is stored in the table.
2. VAR defines a variable and _____ returns the result.
3. YOY stands for _____-over-Year.
4. DATESINPERIOD can return the last _____ days.
5. Filter context applies _____ row context.

Exercise answers | DEFINITION TO REMEMBER

- Match: 1-B, 2-D, 3-C, 4-A, 5-E
- Fill: 1. calculated | 2. RETURN | 3. Year | 4. 30 | 5. before

Module 4 - Visualizations and Reports

Syllabus Table (Blueprint)

Topic	What you must know	Priority
Interface	Visualizations pane, Fields/Data pane, Report canvas	HIGH
Table and charts	Column, bar, line, area, pie, donut, treemap	HIGH
Field wells	Legend, axis, values, tooltips, details	HIGH
KPIs	Card, multi-row card, gauge, KPI visual	MED
Slicers and filters	Slicers, Edit Interactions, sync, Filters pane	HIGH
Buttons and bookmarks	Actions, bookmark states	HIGH
Hierarchies and drill	Drill down vs drill through	HIGH
Grouping	Group, Selection pane, layer order	MED

4.1 Interface and Basics

Item	Where / what
Visualizations pane	Right side - chart types, field wells, Format tab
Fields (Data) pane	Far right - lists data fields (columns)
Report canvas / page	Main area where visuals are placed
Save	File (upper-left) > Save > name the report > Save

Field wells | DEFINITION TO REMEMBER

- Sections where fields are placed: **Legend** = colour coding/grouping; **X/Y axis** = data points; **Values** = numeric calculation (default **Sum**); **Tooltips** = extra info on hover; **Details** = more granular data.

4.2 Charts

Column vs Bar axes | VERY IMPORTANT - HIGH-CHANCE MCQ

- Column chart** (vertical): **X-axis = categories**, **Y-axis = values**.
- Bar chart** (horizontal): **X-axis = values**, **Y-axis = categories**.
- 6 types: **Stacked bar**, **Stacked column**, **Clustered bar**, **Clustered column**, **100% stacked bar**, **100% stacked column**. 100% charts show percentages (in tooltips).
- Clustered column steps: Visualizations pane > clustered column button > drag category to X axis, order total to Y axis (sum is automatic).
- Table** visual: exact figures; resize by dragging; sort by clicking a column header; auto totals at bottom; format via **Format tab**: **Style presets**, **Grid**, **Column header**.
- Line chart**: trends over time; drag order date to X axis, order total to Y axis; date drop-down arrow = year, quarter, month, day.
- Area chart**: line with filled area; **stacked area** shows individual values and total. **Paintbrush icon** opens the Format tab.
- Pie chart**: proportions; payment method to **Legend**, order total to **Values**. **Donut** = ring-shaped pie (swap via donut icon).
- Treemap**: hierarchical data as **nested rectangles**; wells: **Category**, **Details**, **Values**; no separate legend; turn on **Data Labels** in Format tab.

Pie vs Treemap vs Donut | COMMON CONFUSION EXPLAINED

- Pie/Donut = circles, best with few categories. Treemap = rectangles, handles more data points and uses space efficiently.

4.3 KPI Visuals

KPI visuals | VERY IMPORTANT - HIGH-CHANCE MCQ

- **KPI** = Key Performance Indicator aligned with strategic objectives. **Card** = one value. **Multi-row card** = several data points, one per row. **Radial gauge** = progress toward target (default max = double the actual value).
- **KPI visual** wells: **Indicator** (primary measure), **Trend axis** (performance over time), **Target goals** (benchmark).

4.4 Slicers, Filters, Sorting

- **Slicer** = visual filter (list for text, date range for dates); Visualizations pane > Slicer icon > pick field (e.g. Region).
- **Edit Interactions** (Format option in ribbon) controls which visuals a slicer affects; **None icon** disconnects.
- **Sync slicers**: **View menu** > Sync slicers > Advanced options > same **group name** > Sync field changes + Sync filter changes.
- **Sorting**: three dots (top right of visual) > **Sort axis** > field > Sort ascending/descending.
- **Filters pane** scopes: **Filters on this visual / this page / all pages**; remove with **X icon**.

4.5 Buttons and Bookmarks

Buttons | VERY IMPORTANT - HIGH-CHANCE MCQ

- Insert tab > **Buttons** dropdown (right arrow, left arrow, **Clear all slicers...**). Select button > Format pane > **Action** On > type (e.g. **Page navigation**) > destination.
- 8 actions: **Back, Bookmark, Drillthrough, Page navigation, Q&A, Apply all slicers, Clear all slicers, Web URL**. Test with **Ctrl + click**.

Bookmarks | VERY IMPORTANT - HIGH-CHANCE MCQ

- View ribbon > **Bookmarks** > **Add**; rename with three dots. Saves **data properties** (filters/slicers), **display properties** (highlighting/visibility) and **current page** (toggle).
- Bookmark = saved report state; slicers filter live but do not save state.

4.6 Hierarchies, Drill and Grouping

Drill down vs Drill through | COMMON CONFUSION EXPLAINED

- **Drill down** = broader to detailed level **within the same visual** (turn on Drill mode with the drill down icon; drill up icon returns).
- **Drill through** = go to a **separate detail page** filtered to your selection (right-click point > Drill through). Set up by dragging the field into the **Drill through** well; a **Back button** is added.
- **Hierarchy**: Power BI auto-creates **date hierarchy** (Year, Quarter, Month, Day). Custom: right-click field > **Create Hierarchy**, then **Add to Hierarchy**.

Grouping and layering | DEFINITION TO REMEMBER

- Hold **Ctrl**, select visuals > **Format tab > Group**. **Selection pane** = **View tab > Selection**; **Layer Order** tab sets layering (upward arrow moves up). Text box = **Insert tab > Text box**.

Module 4 - Quick Cheat Sheet

- Format tab = Visualizations pane. Sync slicers = View menu. Clear all slicers = Insert > Buttons. Selection pane = View tab.
- Column: X categories, Y values. Bar: X values, Y categories. KPI wells: Indicator, Trend axis, Target goals.

Practice MCQs - Module 4

Q1. Where is the Format tab located?

- (A) Visualizations pane
- (B) Fields pane
- (C) Power Query Home tab
- (D) Model view

Q2. In a column chart the X-axis shows:

- (A) Legend
- (B) Values
- (C) Categories
- (D) Tooltips

Q3. In a bar chart the Y-axis shows:

- (A) Percentages
- (B) Values
- (C) Categories
- (D) Targets

Q4. Which well colour-codes or groups chart items?

- (A) Legend
- (B) Trend axis
- (C) Details
- (D) Values

Q5. A donut chart is:

- (A) A line with fill
- (B) A KPI
- (C) A hierarchical nested-rectangle chart
- (D) A ring-shaped pie chart

Q6. KPI visual field wells are:

- (A) Category, Details, Values
- (B) Legend, Axis, Values
- (C) Rows, Columns, Values
- (D) Indicator, Trend axis, Target goals

Q7. Sync slicers is found under:

- (A) View menu
- (B) Format tab
- (C) Insert tab
- (D) Home tab

Q8. The Clear all slicers button is added from:

- (A) File > Save
- (B) Format tab > Group
- (C) View > Bookmarks
- (D) Insert tab > Buttons dropdown

Q9. Drill through navigates to:

- (A) A bookmark only
- (B) A separate detailed report page
- (C) The same visual at deeper level
- (D) Power Query

Q10. A bookmark saves:

- (A) Only chart colours
- (B) Only the data model
- (C) Filters/slicers, visibility and page state
- (D) Power Query steps

Answer Key - Module 4

Q	Ans	Answer and why
Q1	A	Visualizations pane - Paintbrush icon opens it.
Q2	C	Categories - Bar chart is reversed.
Q3	C	Categories - Bar: X = values.
Q4	A	Legend - Legend well controls grouping.
Q5	D	A ring-shaped pie chart - Treemap uses rectangles.
Q6	D	Indicator, Trend axis, Target goals - These three.
Q7	A	View menu - Then Advanced options > group name.
Q8	D	Insert tab > Buttons dropdown - It resets all slicers.
Q9	B	A separate detailed report page - Drill down stays in the same visual.
Q10	C	Filters/slicers, visibility and page state - Data, display and current page.

Exercise - Match the Following

Column A	Column B
1. Treemap	A. Controls slicer impact on visuals
2. Radial gauge	B. Progress toward a target
3. Edit Interactions	C. Year, Quarter, Month, Day
4. Layer Order	D. Nested rectangles for hierarchical data
5. Date hierarchy	E. Tab in Selection pane

Exercise - Fill in the Blanks

1. Values well performs a _____ by default.
2. Filters pane scopes: this visual, this page and _____ pages.
3. Open Selection pane from the _____ tab.
4. Drill up icon returns to the _____ chart.
5. Bookmark action presents the page linked to the _____.

Exercise answers | DEFINITION TO REMEMBER

- Match: 1-D, 2-B, 3-A, 4-E, 5-C
- Fill: 1. Sum | 2. all | 3. View | 4. main | 5. bookmark

Final Cheat Sheet and Exam Tips

Memorise these 5 | VERY IMPORTANT - HIGH-CHANCE MCQ

- **Storage modes**: Import = in-memory; DirectQuery = real-time; Dual = both (service picks). DirectQuery/Dual to Import is irreversible.
- **Schemas**: Flat = one table; Star = fact + dimensions; Snowflake = normalized dimensions.
- **CALCULATE** modifies filter context; **Row context** = current row; **Filter context** = filters before evaluation.
- **UI paths**: Format tab > Visualizations pane; Sync slicers > View menu; Clear all slicers > Insert > Buttons; New Measure > Table Tools > Calculations group.
- **Joins**: Left outer = all left + matching right; Right outer = all right + matching left; Full outer = all; Inner = only matching.

Top confusions | COMMON CONFUSION EXPLAINED

- Append (rows) vs Merge (columns). Pivot (narrow to wide) vs Unpivot (wide to narrow). Null (numeric/date) vs Empty (text). Unique (once) vs Distinct (different).
- Fact (metrics, centre) vs Dimension (descriptive). Measure (dynamic) vs Calculated column (static). Quick (no code) vs Custom (DAX).
- Drill down (same visual) vs Drill through (new page). Bookmark (saves state) vs Slicer (live filter). Column vs Bar axes are reversed.

Exam tips | DEFINITION TO REMEMBER

- Short, basic MCQs: read every option; keyword-match against the highlighted words in this guide.
- Menu-path questions: remember the TAB name (Home, Transform, View, Insert, Format, Table Tools).
- Do not leave blanks; eliminate two options first. Manage time - shifts are 1 hour each.

Critical Facts - Quick Reference Tables

Module	Critical fact
M1	Import = memory, quick; DirectQuery = real-time; Dual = both; composite model = multiple modes/sources
M1	Flat files: CSV, TXT, Fixed width, Excel. Relational: SQL Server, MySQL, Oracle. NoSQL: MongoDB, Cassandra
M1	Schedule Refresh: Service > semantic model > Refresh > Schedule Refresh. Trigger starts; action performs
M2	Applied Steps right pane; X removes step + later ones; Gear modifies
M2	Append = same columns/names/types; Merge = shared column; 4 join types
M2	Profiling: Column quality, Column distribution, Column profile (View tab > Data Preview)
M2	Star = fact + dimensions; Snowflake = normalized dimensions; Flat = one table; filter direction default Single
M3	Measures dynamic; calculated columns static; CALCULATE changes filter context; KEEPFILTERS/CROSSFILTER/USERRELATIONSHIP
M3	TOTALYTD, DATESYTD, DATESBETWEEN, DATESINPERIOD, SAMEPERIODLASTYEAR, DATEADD, PARALLELPERIOD
M3	Statistical: AVERAGE, MEDIAN, COUNT, DISTINCTCOUNT, MIN, MAX. Quick measure categories = 6
M4	Column: X cat / Y values; Bar: X values / Y cat. Six bar/column types. KPI: Indicator, Trend axis, Target goals
M4	Sync slicers = View; Clear all slicers = Insert > Buttons; Selection pane = View; Group = Format tab
M4	Button actions (8); Bookmark states (3); Filter scopes (3); Date hierarchy: Year, Quarter, Month, Day

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Bonus - Excel and PowerPoint Shortcuts

Excel shortcut	Action
Ctrl + C / X / V	Copy / Cut / Paste
Ctrl + Z / Y	Undo / Redo
Ctrl + S F12	Save Save As
Ctrl + A	Select all
Ctrl + F Ctrl + H	Find Replace
Ctrl + Arrow keys	Jump to edge of data
Ctrl + Shift + L	Toggle filter
Alt + =	AutoSum
F2	Edit cell
F4	Toggle absolute reference (\$) / repeat last action
Ctrl + 1	Format Cells
Ctrl + T	Create table
Ctrl + Home	Go to A1
Ctrl + ;	Insert today's date
Alt + Enter	New line inside a cell
Ctrl + Space / Shift + Space	Select column / row
Ctrl + Shift + 4 / 5	Currency / Percentage format

PowerPoint shortcut	Action
F5	Start slideshow from beginning
Shift + F5	Start from current slide
Esc	End slideshow
Ctrl + M	New slide
Ctrl + D	Duplicate selected object/slide
Ctrl + B / I / U	Bold / Italic / Underline
Ctrl + G / Ctrl + Shift + G	Group / Ungroup
Ctrl + K	Insert hyperlink
B W (in show)	Black screen White screen
Ctrl + Shift + C / V	Copy / Paste formatting
F7	Spelling check

Mock Exam Drill - 50 Questions

Mixed from all four modules. Target: 40+ correct. Answers follow after the questions.

Questions

Q1. ETL stage that collects data from sources is:

- (A) Load
- (B) Extract
- (C) Model
- (D) Transform

Q2. Which mode retrieves metadata, column names and data types only?

- (A) DirectQuery
- (B) Import
- (C) Dual
- (D) Composite

Q3. Which is a relational data source?

- (A) SQL Server
- (B) MongoDB
- (C) Azure Blob Storage
- (D) Cassandra

Q4. Cassandra is an example of:

- (A) Flat file
- (B) Visualization
- (C) Relational database
- (D) NoSQL database

Q5. Images, audio and video are typically:

- (A) Unstructured data
- (B) Structured data
- (C) Relational data
- (D) Fixed width data

Q6. How many connectors does Power BI offer (approx.)?

- (A) 100+
- (B) Exactly 50
- (C) About 10
- (D) About 25

Q7. In Model view, a table header with no colour indicates:

- (A) Calculated table
- (B) Dual
- (C) DirectQuery
- (D) Import mode

Q8. Where is the Storage Mode dropdown?

- (A) Home tab > Get Data
- (B) Format tab
- (C) Properties pane > Advanced section
- (D) Selection pane

Q9. An event or condition that starts a workflow is a:

- (A) Connector
- (B) Hierarchy
- (C) Action
- (D) Trigger

Q10. Another name for a dataset in Power BI Service:

- (A) Semantic model
- (B) Connector
- (C) Report
- (D) Workspace

Q11. Which sign-in option is NOT listed for SQL connection?

- (A) Windows account
- (B) Social media account
- (C) Database credentials
- (D) Microsoft account (Azure AD)

Q12. Which tab holds the Transform Data button to open Power Query Editor?

- (A) Home
- (B) Insert
- (C) View
- (D) Format

Q13. Power Query Editor ribbon tabs include:

- (A) Model, Data, Report, Visual
- (B) Home, Insert, Format, Bookmarks
- (C) Home, Transform, Add Column, View
- (D) File, Edit, Data, Report

Q14. Gear icon next to an applied step:

- (A) Renames the step
- (B) Deletes the step
- (C) Reorders the step
- (D) Opens settings to modify the step

Q15. Use First Row as Headers is in:

- (A) Insert tab
- (B) Format tab
- (C) View tab
- (D) Home tab > Transform group

Q16. Which operation turns rows into columns?

- (A) Pivot
- (B) Append
- (C) Normalize
- (D) Unpivot

Q17. Pivot Column aggregate example used in the course:

- (A) Distinct only
- (B) Count All
- (C) Mode
- (D) Median

Q18. Merge uses:

- (A) Same column names only
- (B) Only Append Queries as New
- (C) The Data pane
- (D) A shared join key column

Q19. Which join returns all records from both tables?

- (A) Inner
- (B) Full outer
- (C) Left outer
- (D) Right outer

Q20. Which join returns only matching rows?

- (A) Right outer
- (B) Inner
- (C) Full outer
- (D) Left outer

Q21. Column quality shows:

- (A) Join keys
- (B) Valid, error and empty rows
- (C) Min and max
- (D) Table relationships

Q22. Standard deviation measures:

- (A) Variation or dispersion
- (B) Row count
- (C) Most frequent value
- (D) Smallest value

Q23. Snowflake schema is:

- (A) A DAX function
- (B) A single table
- (C) A report layout
- (D) Star schema with normalized dimension tables

Q24. Fact tables contain:

- (A) Slicers
- (B) Descriptive text only
- (C) Quantifiable metrics of a business process
- (D) Bookmarks

Q25. Typical cardinality in star schema relationships:

- (A) None
- (B) One-to-many (many to one)
- (C) One-to-one only
- (D) Many-to-many only

Q26. Which setting is disabled before manually building relationships?

- (A) Replace Values
- (B) Close and Apply
- (C) Data profiling
- (D) Auto detect new relationships

Q27. DAX stands for:

- (A) Dynamic Analysis XML
- (B) Data Aggregate Exchange
- (C) Data Analysis Expressions
- (D) Direct Access Queries

Q28. A measure is calculated:

- (A) Only for text
- (B) Only in Power Query
- (C) In the context of the visualization
- (D) Once, stored in the table

Q29. SUMX does what?

- (A) Shifts dates
- (B) Iterates rows, multiplies and sums
- (C) Counts distinct values
- (D) Returns the median

Q30. TOPN is used to:

- (A) Create a table
- (B) Define top N products
- (C) Keep filters
- (D) Shift dates

Q31. USERELATIONSHIP is a CALCULATE modifier that:

- (A) Sorts the axis
- (B) Sums the year
- (C) Keeps existing filters
- (D) Specifies which relationship to use

Q32. Filter context is applied:

- (A) Before row context
- (B) Never with CALCULATE
- (C) After row context
- (D) Only in Power Query

Q33. DATESBETWEEN defines:

- (A) Year to date
- (B) Last 30 days
- (C) Same period last year
- (D) A date range between two dates

Q34. DATESINPERIOD can return:

- (A) Distinct values
- (B) Dates in a period such as the last 30 days
- (C) Dates from Jan 1
- (D) Only the current year

Q35. DATEADD example in the course shifts revenue by:

- (A) Two years back
- (B) One quarter forward
- (C) One month back
- (D) Zero days

Q36. PARALLELPERIOD example shifted order date by:

- (A) Thirty days
- (B) Two years
- (C) One month
- (D) One week

Q37. Which function returns the middle value of a set?

- (A) MIN
- (B) MEDIAN
- (C) COUNT
- (D) AVERAGE

Q38. Predictive analytics:

- (A) Creates relationships
- (B) Forecasts future trends from current/historical data
- (C) Cleans data
- (D) Draws charts only

Q39. The Fields (Data) pane lists:

- (A) Data fields corresponding to columns
- (B) Applied steps
- (C) Chart types
- (D) Bookmarks

Q40. Default calculation when a numeric field is dropped in Values:

- (A) Average
- (B) Count
- (C) Median
- (D) Sum

Q41. Which chart shows hierarchical data as nested rectangles?

- (A) Line
- (B) Treemap
- (C) Donut
- (D) Area

Q42. Radial gauge default max value is:

- (A) Double the actual value
- (B) Zero
- (C) 100 always
- (D) Half the actual value

Q43. Edit Interactions is used to:

- (A) Create hierarchies
- (B) Group visuals
- (C) Control which visuals a slicer affects
- (D) Save the report

Q44. Sync slicers group names must:

- (A) Be left blank
- (B) Match exactly
- (C) Be different
- (D) Be numeric

Q45. Page navigation action is set in:

- (A) Filters pane
- (B) Model view
- (C) Power Query Editor
- (D) Format pane > Action section

Q46. Which is NOT a button action?

- (A) Q&A
- (B) Create hierarchy
- (C) Back
- (D) Web URL

Q47. Auto-created date hierarchy levels are:

- (A) Category, Subcategory, Name
- (B) Week, Hour, Minute
- (C) Decade, Year, Day
- (D) Year, Quarter, Month, Day

Q48. Filters on all pages applies filters to:

- (A) Every page in a multi-page report
- (B) Slicers only
- (C) Current page only
- (D) Selected visual only

Q49. Selection pane shows:

- (A) Applied steps
- (B) DAX measures
- (C) All visuals and groups with Layer Order
- (D) Relationships

Q50. How do you add a report title text?

- (A) Insert tab > Text box
- (B) View tab > Bookmarks
- (C) Format tab > Grid
- (D) Home tab > Merge

Mock Drill - Answer Key

Q	Ans	Answer and why
Q1	B	Extract - First stage.
Q2	A	DirectQuery - Data stays at the source.
Q3	A	SQL Server - Also MySQL and Oracle.
Q4	D	NoSQL database - Also MongoDB.
Q5	A	Unstructured data - No predefined format.
Q6	A	100+ - Free to use.
Q7	D	Import mode - Same colour = DirectQuery.
Q8	C	Properties pane > Advanced section - Right-click table > Properties.
Q9	D	Trigger - Actions are the tasks.
Q10	A	Semantic model - Same thing.
Q11	B	Social media account - Three options only.
Q12	A	Home - Queries group.
Q13	C	Home, Transform, Add Column, View - Four tabs.
Q14	D	Opens settings to modify the step - X deletes.
Q15	D	Home tab > Transform group - Promotes first row.
Q16	A	Pivot - Narrow to wide.
Q17	B	Count All - Advanced options.
Q18	D	A shared join key column - Consolidates tables.
Q19	B	Full outer - Regardless of matching.
Q20	B	Inner - Both sides match.
Q21	B	Valid, error and empty rows - Per column.
Q22	A	Variation or dispersion - Used with mean.
Q23	D	Star schema with normalized dimension tables - More tables.
Q24	C	Quantifiable metrics of a business process - Order ID, Quantity, Total Price.
Q25	B	One-to-many (many to one) - Dimension to fact.
Q26	D	Auto detect new relationships - Options > Data Load.
Q27	C	Data Analysis Expressions - Power BI formula language.
Q28	C	In the context of the visualization - Dynamic.
Q29	B	Iterates rows, multiplies and sums - e.g. quantity x unit price.
Q30	B	Define top N products - Top Two Products measure.
Q31	D	Specifies which relationship to use - Others: KEEPFILTERS, CROSSFILTER.
Q32	A	Before row context - Order matters.
Q33	D	A date range between two dates - SummerSales example.
Q34	B	Dates in a period such as the last 30 days - Last30DaysSales.
Q35	C	One month back - Month-over-month comparison.
Q36	B	Two years - Compare with two years ago.
Q37	B	MEDIAN - Statistical function.
Q38	B	Forecasts future trends from current/historical data - BI concept.
Q39	A	Data fields corresponding to columns - Far right.

Q	Ans	Answer and why
Q40	D	Sum - Sum of order total.
Q41	B	Treemap - Uses Category, Details, Values wells.
Q42	A	Double the actual value - Set target min/max for realism.
Q43	C	Control which visuals a slicer affects - None icon disconnects.
Q44	B	Match exactly - Same group name on both pages.
Q45	D	Format pane > Action section - Set Action to On first.
Q46	B	Create hierarchy - Eight actions total.
Q47	D	Year, Quarter, Month, Day - Power BI creates for date fields.
Q48	A	Every page in a multi-page report - Drag field into that section.
Q49	C	All visuals and groups with Layer Order - View tab > Selection.
Q50	A	Insert tab > Text box - Then change font size.

Score yourself | VERY IMPORTANT - HIGH-CHANCE MCQ

- 45-50: Exam ready. 38-44: Re-read the red and teal boxes. Below 38: Re-read each module, then redo the module MCQs.

Bonus extra: **Important concepts in influence** - Storage modes, Schemas, Context (row/filter), Append vs Merge, Drill down vs Drill through, and menu locations (View, Insert, Format, Table Tools). These repeat the most across the course.

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ALL THE BEST!

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